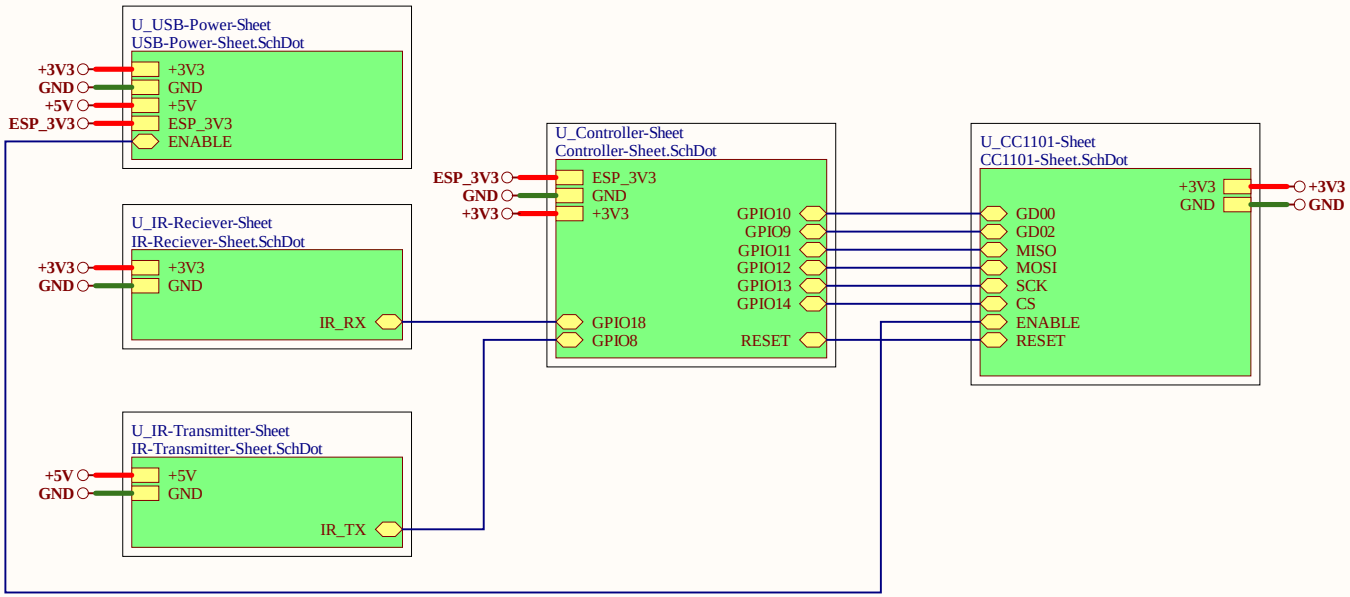
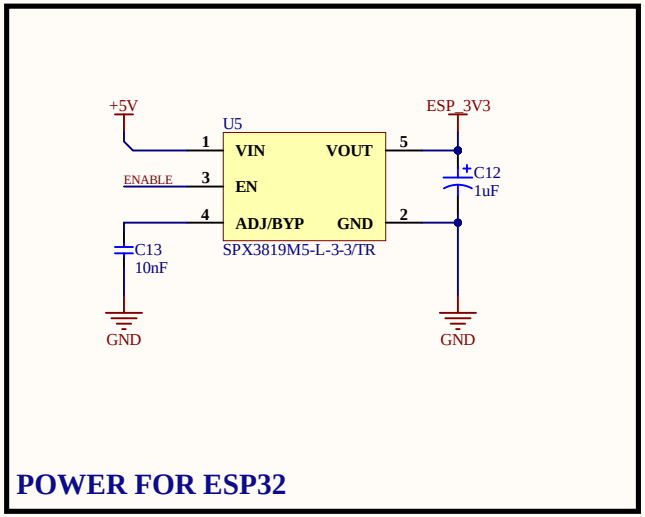
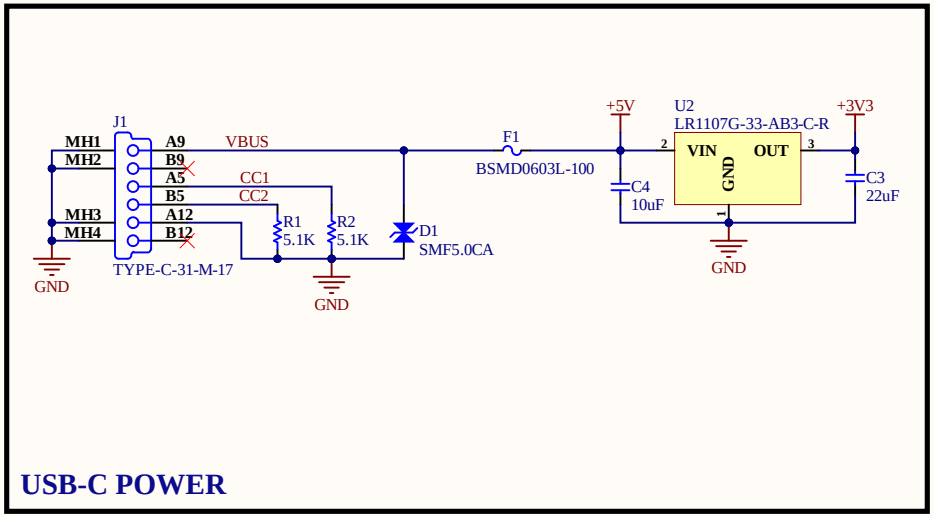


Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Top-Sheet.SchDot	
File: RF-IR-Blaster-v1.1.PrjPcb	
Date: 12/29/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 6



POWER FOR ESP32



USB-C POWER



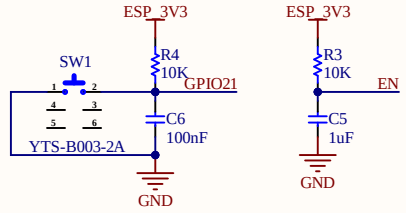
TEST POINTS



Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: USB-Power-Sheet.SchDot	
File: RF-IR-Blaster-v1.1.PrjPcb	
Date: 12/29/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 6

BOOT



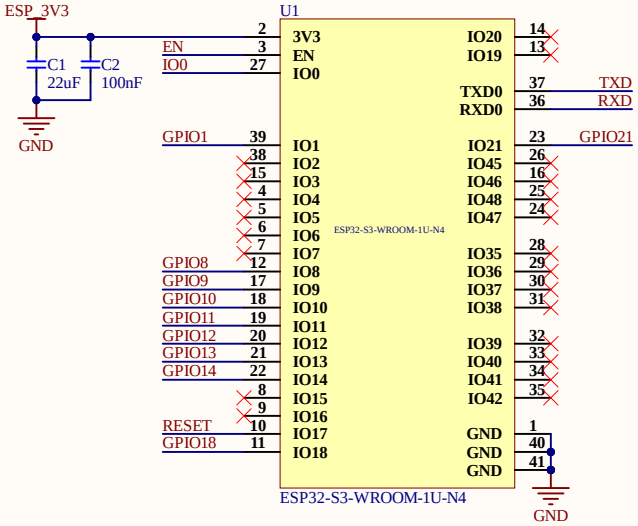
Strapping pins
• BOOT = IO0
• JTAG = IO3
• VSPI = IO45
• LOG = IO46

SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses.
When using Octal Flash or Octal PSRAM or both, GPIO3337 are connected to SPIIO4 SPIIO7 and
SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO3337 are also not recommended for
other uses.

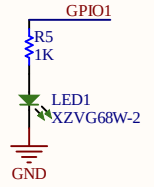
For 2MB PSRAM version (embedded chip ESP32-S3R2 quad SPI) pins IO35, IO36, IO37 should be free to use.
For 8MB PSRAM version (embedded chip ESP32-S3R8 octal SPI) pins IO35, IO36, IO37 connect to PSRAM and
are not available for other uses.

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs,
USB-JTAG will be disabled by the drivers.

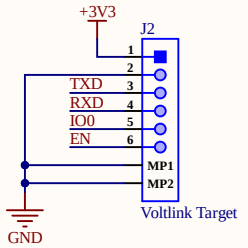
MCU



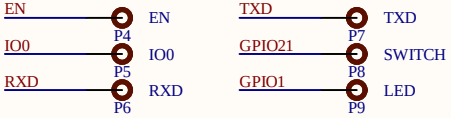
INDICATION

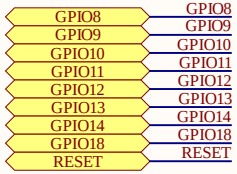


UART



TEST POINTS

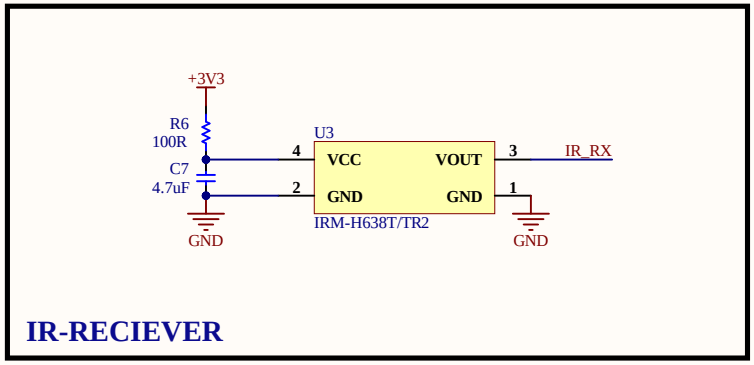






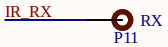
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Controller-Sheet.SchDot	
File: RF-IR-Blaster-v1.1.PrjPcb	
Date: 12/29/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6

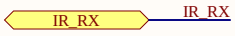


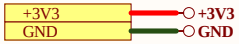
IR-RECIEVER

Everlight Elec IRM-H638T/TR2
NEC YES
RC5_Philips YES
RC6_Philips YES
RCA_Thomson YES
Toshiba YES
SHarp YES
Sony 12Bit YES
Sony 15Bit NO
Sony 20Bit NO
Matushita YES
Mitsubishi NO
Zenith YES
JVC NO
Continous Code NO
High Data Rate Code NO
*RC6 is only compatible if the data low time is 25ms or more.



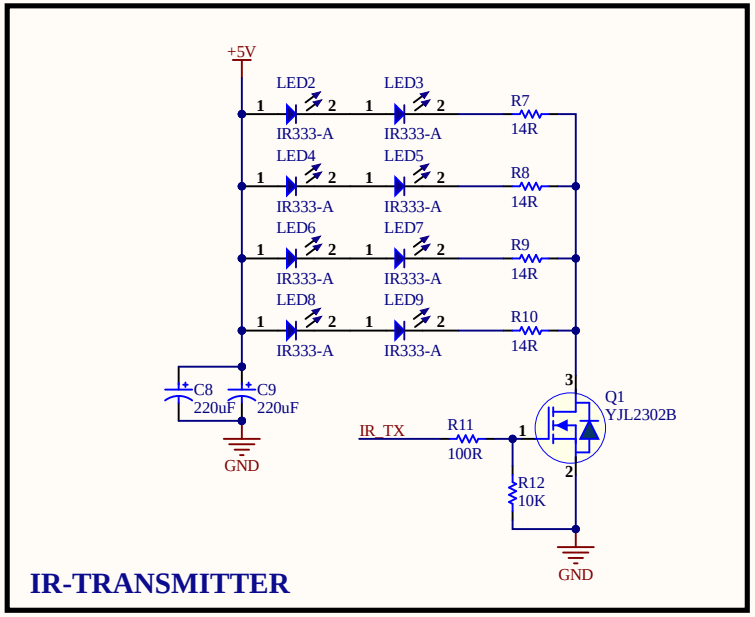
TEST POINTS



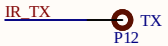


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: IR-Reciever-Sheet.SchDot	
File: RF-IR-Blaster-v1.1.PrjPcb	
Date: 12/29/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 6

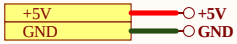


940nm If=100mA Vf=1.4V
IR333-A has an angle of 20deg
IR333-C has an angle of 30deg



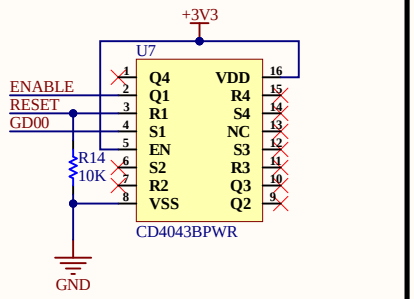
TEST POINTS



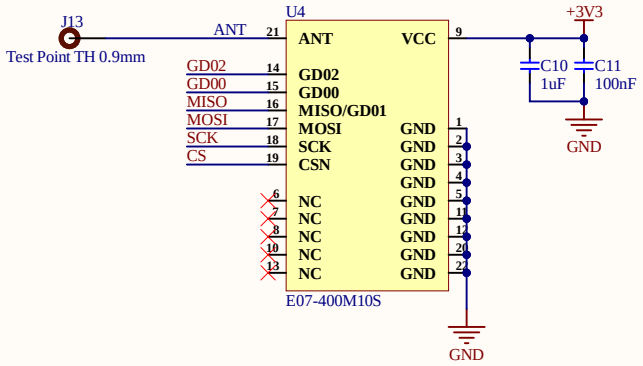


Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: IR-Transmitter-Sheet.SchDot	
File: RF-IR-Blaster-v1.1.PrjPcb	
Date: 12/29/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 6

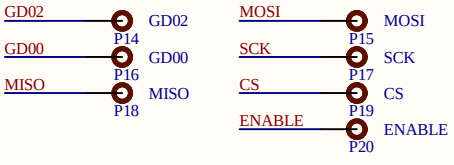


SR Latch

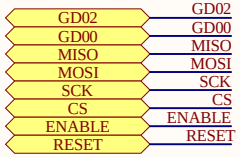


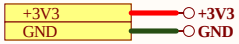
CC1101

EBYTE module are available in two variants 400MHz and 900MHz
E07-900M10S(has CE and FCC) E07-400M10S(just CE)
For future revisions we should consider implementing our own RF section based on CC1101 or the newer CC110L



TEST POINTS



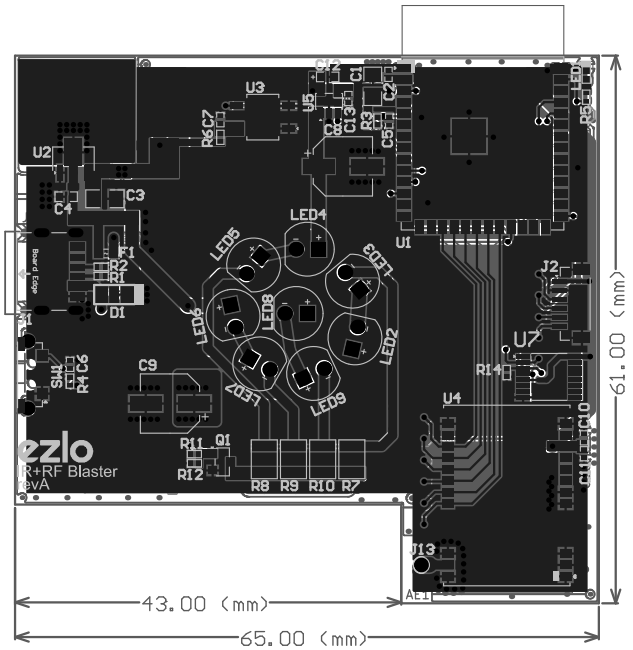


Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: CC1101-Sheet.SchDot

File: RF-IR-Blaster-v1.1.PrjPcb

Date: 12/29/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.39mil	3.3	
1	Top Layer		1.38mil		
	Dielectric 1	FR-4	59.45mil	4.5	
2	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.39mil	3.3	
	Bottom Overlay				
Total board thickness:			62.99mil		